Vivekanand Education Society's Institute of Technology

Department of Computer Engineering

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Project Synopsis (2025-26) - Sem VII

NeuroSplain: Interpretable Multi-Modal Framework for Alzheimer's Disease Detection

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Abstract:

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder that affects memory, cognition, and the ability to perform daily activities. As the disease advances, it places a significant emotional, physical, and financial strain on patients, caregivers, and healthcare systems. Early detection is vital for improving patient quality of life, enabling timely treatment, and providing opportunities for lifestyle adjustments that may slow disease progression. However, current diagnostic approaches often combine cognitive tests, patient history, and biomarker analysis, which can be time-consuming, resource-intensive, and sometimes inconsistent across different healthcare settings.

This project proposes **NeuroSplain**, a multi-technique Explainable AI (XAI) system designed for Alzheimer’s detection using non-imaging clinical, lifestyle, and biomarker features. The system integrates data from kaggle dataset as well as our custom generated data, applies robust preprocessing, and uses feature selection to identify the most relevant predictors. A combination of ante-hoc and post-hoc XAI techniques, like SHAP, LIME, and feature importance analysis, ensures that predictions are not only accurate but also interpretable for clinicians. By pairing strong predictive performance with clear explanations, NeuroSplain aims to build clinician trust, support data-driven decision-making, and contribute to earlier, more effective management of Alzheimer’s disease.

Introduction:

Alzheimer’s disease is one of the most common causes of dementia, gradually impairing memory, thinking ability, and daily functioning [1]. It currently affects millions worldwide, and its prevalence continues to grow with the aging global population. The personal and societal impacts are immense as patients face a progressive loss of independence, while caregivers and healthcare systems bear increasing emotional, physical, and financial burdens. Early detection is essential, as timely diagnosis allows for better disease management, early intervention strategies, and improved patient quality of life. However, conventional diagnostic approaches often rely on lengthy cognitive assessments, in-person evaluations, and costly laboratory or imaging tests, which may delay identification of the disease in its earlier, more treatable stages.

Artificial Intelligence (AI) has shown great potential in enhancing the speed and accuracy of Alzheimer's detection by identifying complex patterns in patient data that may not be obvious to human clinicians. Yet, despite its promise, AI adoption in clinical settings remains limited due to the "black box" nature of many models. Explainable AI (XAI) addresses this barrier by providing transparent reasoning behind predictions, fostering trust, and enabling medical experts to validate outputs against clinical knowledge [2]. The NeuroSplain system leverages multi-technique XAI applied to diverse, non-imaging data sources such as demographics, lifestyle factors, biomarkers, and cognitive scores within a structured architecture covering data acquisition, feature analysis, model building, and interpretation.

Problem Statement:

Alzheimer's disease presents unique challenges for early detection due to the complexity of its causes and symptoms, as well as the diverse nature of patient data. While Artificial Intelligence (AI) has shown great promise in healthcare diagnostics, its adoption in Alzheimer's detection is hindered by data fragmentation, limited explainability, and dependence on high-cost imaging. Clinical and lifestyle data, which are more accessible, remain underutilized despite their potential to reveal strong predictive patterns when combined effectively. A robust, explainable AI system that can integrate these varied data sources is crucial to improving early diagnosis, increasing clinical trust, and making such solutions more accessible in real-world healthcare settings [3].

Challenges in Current Approaches:

- Fragmented and heterogeneous patient data across sources.
- Limited use of explainability in AI models for clinical decision support.
- Over-reliance on imaging data, which is expensive and less accessible than clinical and lifestyle records.
- Difficulty in integrating multiple feature types (demographics, lifestyle, biomarkers, cognitive assessments) into a single robust detection system.

Proposed Solution:

The proposed solution aims to develop a **multi-technique Explainable AI (XAI) system** for early and accurate detection of Alzheimer's disease using non-imaging clinical and lifestyle data. The system will integrate multiple preprocessing steps, including missing value handling, categorical encoding, and normalization, followed by advanced feature selection methods such as correlation-based filtering and mutual information analysis to retain only the most relevant predictors [6]. The final predictions will be explained through interpretable frameworks such as SHAP, LIME, and counterfactual explanations to ensure transparency for clinicians.

In addition, the solution emphasizes **clinical applicability and decision support** by providing user-friendly visualizations of model explanations, highlighting the influence of features like cognitive test scores, age, lifestyle factors, and genetic markers on predictions. This approach not only improves diagnostic accuracy but also bridges the trust gap between AI systems and healthcare professionals by offering insights into the decision-making process. The integration of multiple XAI methods ensures a more holistic and reliable interpretation, which can assist in personalized patient care and facilitate early interventions [6].

Methodology / Block Diagram :

Methodology :

The NeuroSplain system will be implemented through a structured, four-phase workflow, as illustrated in the block diagram.

1. Data Acquisition & Preparation

- **Dataset Acquisition & Synthetic Data Generation** – Gather real datasets and generate synthetic samples to enhance diversity and privacy.
- **Data Preprocessing** – Clean, encode, and normalize data while handling missing values.
- **Correlation-Based Feature Analysis** – Remove highly correlated or redundant features to improve efficiency.

2. Feature Engineering & Exploratory Analysis

- **Data Visualization** – Use heatmaps, box plots, scatter plots, and histograms to explore feature relationships.

- **Data Balancing** – Equalize class distribution using SMOTE or undersampling techniques.
- **Feature Selection** – Apply methods like K-Best and Chi-Square to identify top predictive features.

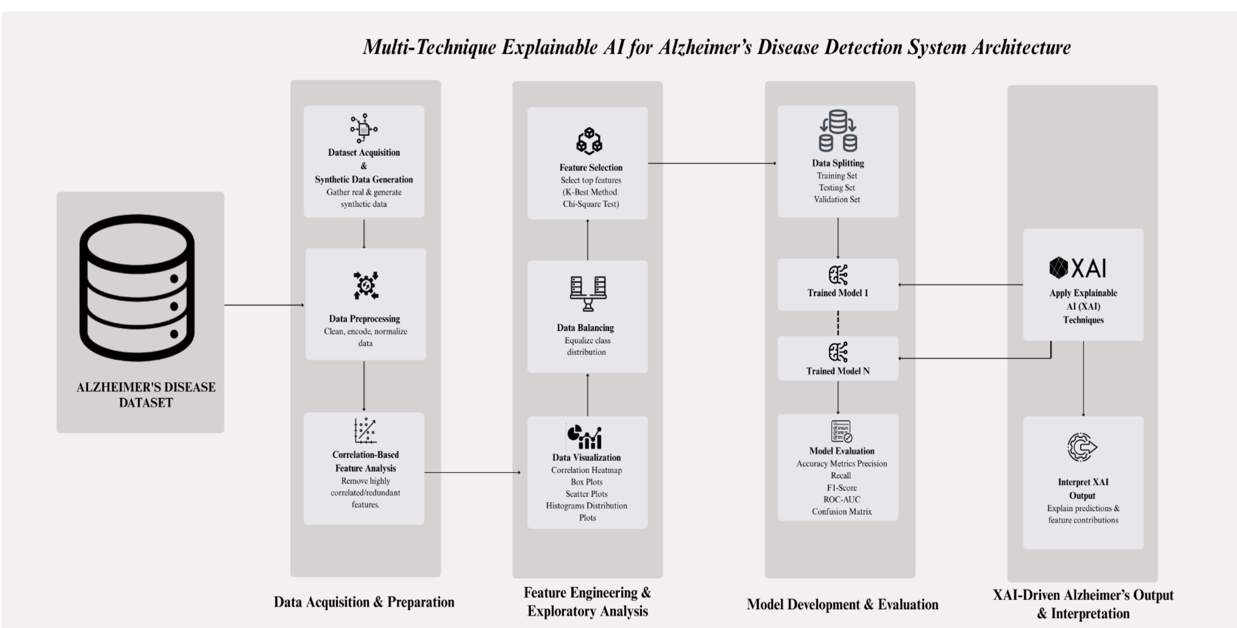
3. Model Development & Evaluation

- **Data Splitting** – Divide the dataset into training, testing, and validation sets.
- **Trained Models** – Build and tune multiple machine learning models (Random Forest, Gradient Boosting, Logistic Regression, etc.).
- **Model Evaluation** – Assess models using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and confusion matrices.

4. XAI-Driven Alzheimer's Output & Interpretation

- **Applying XAI Techniques:** Implement methods such as SHAP, LIME, Feature Importance plots, and Partial Dependence Plots to interpret predictions.
- **Interpretation of Results:** The XAI methods will provide detailed reasoning behind each classification, explaining why the model predicted a particular outcome, which features contributed the most, and how these features influenced the decision.

Block Diagram :



Hardware , Software and tools Requirements:

Hardware required:

Processor	**Intel i5 / AMD Ryzen 5 or higher**
RAM	**Minimum 8 GB (16 GB recommended)**
Storage	**50 GB free space**
Screen Resolution	**1366×768 or higher**

Software required:

Frontend (Visualization & Interface)	**Streamlit / Flask web app**
Backend & Modeling	**Python (Scikit-learn, Pandas, NumPy, Matplotlib, XAI tools)**
Other Tools	**Google Colab**

Proposed Evaluation Measures :

The success of the NeuroSplain system will be determined through a combination of technical, usability, and clinical relevance assessments. The evaluation process will ensure that the model is not only accurate but also interpretable, user-friendly, and aligned with real-world medical needs. Both quantitative metrics and qualitative feedback from clinicians will be considered to validate the system's performance and practicality for Alzheimer's disease detection.

Evaluation Criteria:

- **Model Performance** – Accuracy, Precision, Recall, F1-Score, and ROC-AUC will be used to measure predictive strength.
- **Interpretability** – Quality and clarity of explanations generated by XAI methods.

- **Data Handling** – Robustness in managing missing values, handling class imbalance etc.
- **Clinical Relevance** – Alignment of model findings with established Alzheimer's research and applicability in real-world diagnostic workflows.

Conclusion:

The NeuroSplain project is designed to address one of the most pressing challenges in neurology: early and reliable detection of Alzheimer's disease by combining the power of Artificial Intelligence with the transparency of Explainable AI. By focusing on non-imaging clinical, lifestyle, and biomarker data, the system leverages more accessible and cost-effective sources of information compared to traditional imaging-based approaches. Through rigorous preprocessing, feature selection, multi-model evaluation, and integration of advanced XAI techniques, NeuroSplain ensures that its predictions are both accurate and interpretable, fostering greater confidence among clinicians and stakeholders in healthcare.

This approach not only holds the potential to improve early detection rates but also to enable more personalized care strategies tailored to individual patient profiles. The interpretability offered by XAI bridges the gap between AI outputs and clinical reasoning, making it easier for medical professionals to validate and trust the results. Looking ahead, the system can be expanded to incorporate imaging modalities, integrate seamlessly into hospital IT infrastructures, and undergo continuous retraining as new patient data and medical insights emerge. By doing so, NeuroSplain aims to remain a relevant, adaptable, and impactful tool in the global fight against Alzheimer's disease.

References:

[1] T. Mahmud, K. Barua, S. U. Habiba, N. Sharmen, M. S. Hossain, and K. Andersson, "An explainable AI paradigm for Alzheimer's diagnosis using deep transfer learning," Diagnostics, vol. 14, no. 3, art. no. 345, 2024

[2] V. Viswan, N. Shaffi, M. Mahmud, K. Subramanian, and F. Hajamohideen, "Explainable Artificial Intelligence in Alzheimer's Disease Classification: A Systematic Review," Cognitive Computation, vol. 16, no. 1, pp. 1–44, 2024

[3] M. T. Khosroshahi, S. Morsali, S. Gharakhanlou, A. Motamedi, S. Hassanbaghlou, H. Vahedi, S. Pedrammehr, H. M. Dipu Kabir, and A. Jafarizadeh, "Explainable Artificial

Intelligence in Neuroimaging of Alzheimer's Disease," Diagnostics, vol. 15, no. 5, art. no. 612, Mar. 2025

[4] E. Mahamud, M. A. Assaduzzaman, J. Islam, N. Fahad, M. J. Hossen, and T. T. Thirumalaiappan Ramanathan, "Enhancing Alzheimer's disease detection: An explainable machine learning approach with ensemble techniques," Intelligence-Based Medicine, vol. 11, art. no. 100240, 2025

[5] S. M. Mahim, M. S. Ali, M. O. Hasan, A. A. N. Nafi, A. Sadat, and S. A. Hasan, "Unlocking the potential of XAI for improved Alzheimer's disease detection and classification using a ViT-GRU model," IEEE Access, vol. 12, pp. 14014-14029, 2024

[6] Y. Dubey, A. Bhongade, P. Palsodkar, and P. Fulzele, "Efficient explainable models for Alzheimer's disease classification with feature selection and data balancing approach using ensemble learning," Diagnostics, vol. 14, no. 24, art. no. 2770, 2024